FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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WIA3002 ACADEMIC PROJECT I REPORT

INTERACTIVE STUDENT SPORTS PORTAL WITH HEALTH DASHBOARD

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INTERACTIVE STUDENT SPORTS PORTAL WITH HEALTH DASHBOARD

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REPORT SUBMITTED IN FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF COMPUTER SCIENCE (INFORMATION SYSTEMS)

FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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ACKNOWLEDGEMENTS

I would like to express my deepest gratitude to everyone who contributed to the completion of this project.

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ABSTRACT

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CHAPTER 1: INTRODUCTION TO PROJECT

1.1 PROJECT BACKGROUND

This project is developed in close collaboration with Institut Sukan Negara (ISN) to build the Athlete Injury Risk Management System (AIRMS), a dedicated platform aimed at transforming how sports institutions monitor athlete health, manage injury data, and support medical decision-making. This project continues from the foundational work established by previous iterations of the Student Sports Portal (SSP), developed by Sherwynd Liew and Tan Ke Ying, which focused on sports event management and student participation tracking within educational institutions.

While the existing SSP platform successfully addresses event registration, certification, and announcement management, it lacks a dedicated system for longitudinal injury tracking, workload monitoring, and data-driven risk assessment. ISN, which oversees 52 sports across multiple national programs including elite, development, and competition-based programs, requires a more robust digital infrastructure to support its medical and administrative staff in managing athlete welfare at a professional level.

This new platform addresses those gaps by providing a centralized system where athletes can log their activity and self-report non-training injuries, medical staff can manage official injury records and trace individual athlete histories, and administrators can gain holistic insights into injury trends across sports, demographics, and time periods. The system is designed to operate as a separate deployment from the existing SSP, reflecting the distinct user groups and data sensitivity requirements of a professional sports institution versus a student-facing portal.

A key feature of the platform is its dual-dashboard approach, designed around two distinct user needs identified by Dr. Thung from ISN. The administrative dashboard provides a holistic overview of injury data filterable by sport, gender, age group, body part, and time period, enabling policy-level decision-making. The medical staff dashboard focuses on individual athlete traceability, allowing physios, doctors, and strength and conditioning staff to track an athlete's injury history and workload trends over time. Together these dashboards transform raw injury and activity data into actionable clinical and administrative insights.

1.2 PROJECT STATEMENT

1. **Absence of a Dedicated Injury Tracking and Risk Management System**
   Despite the existence of the SSP platform for event and participation management, there is currently no dedicated system for tracking athlete injuries and assessing injury risk over time. Medical staff at ISN rely on fragmented records and manual processes to monitor athlete health, making it difficult to identify patterns, flag at-risk athletes, or maintain a consistent longitudinal record of each athlete's injury history. This gap in infrastructure reduces the effectiveness of injury prevention programs and delays clinical decision-making. Digital injury surveillance literature identifies this fragmentation as a systemic weakness in institutional sports health management, noting that the most effective systems are those that support structured, multi-stakeholder data capture in a consistent and auditable manner (Costello et al., 2024).

2. **Lack of Workload Monitoring for Athlete Injury Risk Assessment**
   Current systems do not provide any mechanism for monitoring athlete training workload in relation to injury risk. Research has established that the ACWR is a reliable indicator of injury risk, with sudden spikes in training load significantly increasing the likelihood of injury, and that presenting workload data through visually categorized risk thresholds significantly aids practitioners in making timely decisions (Andrade et al., 2020). Without a system that computes and displays this metric, athletes and medical staff have no data-driven basis for identifying dangerous workload patterns before injuries occur.

3. **No Centralized Platform for Administrative Injury Analytics**
   Administrators at ISN currently lack a centralized dashboard for viewing injury data across the full breadth of ISN's programs. The IOC consensus statement on sports injury surveillance establishes that meaningful injury trend analysis requires consistent data structured around recognized variables including injury type, body part, severity, mechanism, and time of occurrence (Bahr et al., 2020). Without a system grounded in these principles, producing summaries for management and policy-level stakeholders remains a manual and unreliable process, limiting ISN's ability to respond proactively to injury trends and demonstrate program outcomes to higher authorities.

1.3 PROJECT OBJECTIVES

1. **To identify the requirements for an athlete injury risk management system through stakeholder collaboration with ISN**
   This objective focuses on gathering detailed system requirements through meetings with Dr. Thung and relevant ISN personnel, supported by a review of established literature on injury risk monitoring, workload management, and sports injury surveillance. The requirements gathered will ensure the system addresses the specific operational needs of ISN's administrative and medical staff across all 52 sports programs, grounded in both stakeholder input and academically validated frameworks.

2. **To develop a centralized platform that integrates athlete activity tracking, injury logging, and role-specific dashboards for injury risk management**
   The platform will be developed as a dedicated system separate from the existing SSP deployment, featuring modules for activity tracking, workload calculation, injury and recovery logging, data import and management, and role-specific dashboards for both administrators and medical staff. The system's data model will be grounded in the IOC consensus variables for injury surveillance, ensuring that captured data is analytically meaningful and supports both clinical and administrative use cases. The system will support three user roles — Athlete, Medical Staff, and Administrator — each with tailored access reflecting their distinct responsibilities.

3. **To evaluate the effectiveness and usability of the system in supporting injury risk monitoring and data-driven decision-making at ISN**
   This objective involves testing the system against the functional requirements established with Dr. Thung and evaluating its usability through stakeholder feedback. Evaluation will assess whether the system successfully enables administrators to generate injury summaries consistent with institutional reporting needs, and whether medical staff can effectively trace individual athlete histories and make informed clinical decisions using the platform.

1.4 PROJECT SCOPE

Despite growing emphasis on athlete welfare and sports science at the institutional level, professional sports organizations like ISN continue to rely on manual and fragmented approaches to injury tracking and risk assessment. The absence of a centralized digital platform means that injury data is stored in disconnected Excel files, administrative reporting is conducted manually, and medical staff have no systematic way to trace an individual athlete's development over time. These shortcomings align with gaps identified in the literature, where traditional paper-based and informal reporting approaches are characterized as prone to inconsistency, underreporting, and fragmentation (Costello et al., 2024).

The Athlete Injury Risk Management System addresses these gaps through six functional modules. On the athlete-facing side, the system allows athletes to log their training activities and self-report non-training injuries, with a dashboard that computes and displays their current injury risk level based on the ACWR methodology established in the workload monitoring literature (Andrade et al., 2020). On the institutional side, medical staff can log official injury records, manage recovery status, and access individual athlete profiles with full injury and workload history. Administrators can view holistic injury analytics across all sports, apply demographic and temporal filters aligned with IOC-recommended data variables (Bahr et al., 2020), and generate PDF reports for management and policy purposes. A dedicated data management module supports the upload and validation of Excel-based screening data from ISN, ensuring the database can be updated efficiently as new assessments are conducted.

The system will be deployed as a standalone platform separate from the existing SSP, reflecting the distinct user base and data confidentiality requirements of ISN's athlete records. All athlete data will be handled in compliance with the Non-Disclosure Agreement signed with ISN.

1.5 PROJECT SCHEDULE

Activities	Week													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Decide on Project Title														
Meeting with Supervisor														
Develop Project Objectives														
Conduct Literature Review														
Decide on Methodology														
Prepare System Analysis and Design														
Stakeholder Meeting														
Modifications (Stakeholder Feedback)														
Report Writing														
System Development														
Monitoring Session														
Modifications (Monitoring Feedback)														
Viva Session														

Table 1.1: Project Schedule Academic Project I

CHAPTER 2: LITERATURE REVIEW

2.1 WORKLOAD MONITORING AND INJURY RISK MANAGEMENT IN STUDENT SPORTS

In the context of sports participation, managing physical workload is a critical yet often overlooked aspect of student wellness. As students engage in recurring sporting activities, the accumulation of training load over time can either support fitness development or, if poorly managed, contribute to injury and burnout. One widely adopted approach to quantifying this relationship is the Acute:Chronic Workload Ratio (ACWR), which compares a participant's recent activity load against their longer-term baseline as a means of identifying elevated injury risk.

The systematic review by Andrade et al. (2020) examined the association between ACWR and time-loss injury risk across professional team sports, synthesizing findings from multiple studies to evaluate how different methodological approaches to calculating and interpreting the ratio affect its predictive validity. The review established that ACWR is a practically applicable metric for flagging periods of disproportionate workload, particularly when sudden spikes in activity occur relative to a participant's established baseline. It further highlighted that presenting workload data visually through thresholds categorized as low, moderate, or high risk significantly aids practitioners in making timely, informed decisions about participant readiness and activity management.

These findings provide a strong academic basis for incorporating workload computation and risk visualization into a student sports portal. A system that derives risk levels from logged activity data and presents them through intuitive visual indicators equips students with personalized, actionable health feedback that extends well beyond simple activity counts, supporting proactive wellness management rather than reactive responses to fatigue or injury.

However, the review acknowledges that the bulk of evidence underpinning ACWR originates from professional sports settings, where structured training programmes and consistent data collection are standard. Translating this methodology to a self-reporting student platform introduces challenges around data consistency, as the accuracy of computed ratios depends heavily on the regularity of user input. In conclusion, while ACWR represents a well-validated framework for workload and risk assessment, its meaningful integration into a student-facing system requires careful attention to how activity data is collected and maintained.

2.2 DIGITAL INJURY SURVEILLANCE AND REPORTING SYSTEMS IN SPORTS MEDICINE

The systematic and accurate recording of sports injuries has long been recognized as fundamental to understanding injury patterns, informing prevention strategies, and guiding clinical decision-making. Traditional paper-based or informal reporting approaches, however, are prone to inconsistency, underreporting, and fragmentation — particularly in institutional settings where athletes, medical staff, and administrators operate across different workflows. Digital injury surveillance systems have emerged as a response to these shortcomings, offering structured, multi-stakeholder platforms for capturing and managing injury data in a consistent and auditable manner.

The scoping review by Costello et al. (2024) surveyed the landscape of sports injury surveillance systems in current practice, examining the methodologies employed, the types of data collected, and the extent to which existing systems support the diverse needs of their users. The review found considerable variability across systems in terms of data granularity, reporting frequency, and the degree to which athlete self-reporting was incorporated alongside clinician-recorded entries. It noted that the most effective surveillance systems were those that supported multiple reporting pathways — allowing athletes to submit their own injury observations while enabling medical staff to review, verify, and formalize those submissions — thereby improving both the volume and quality of captured data.

These findings directly inform the design of a sports portal that accommodates distinct injury recording workflows for athletes and medical personnel. Providing athletes with a self-report pathway that feeds into a clinician review process mirrors the multi-stakeholder surveillance architecture identified as most effective in the literature, while the separation of pending, approved, and rejected report states ensures clinical accountability is maintained throughout. The ability for medical staff to manage official injury records — including recovery status updates and historical tracking per athlete — further aligns with the structured, longitudinal data collection approach endorsed by contemporary surveillance methodology.

The review does, however, surface notable limitations in the field. Many existing surveillance systems remain siloed within specific sports organizations or research programmes, limiting their generalizability and scalability to broader institutional contexts such as university sports environments. Furthermore, the reliance on voluntary reporting from athletes introduces persistent underreporting bias, which no purely digital system can fully eliminate without supplementary engagement mechanisms. In conclusion, digital injury surveillance systems represent a significant advancement over traditional recording practices, and the design principles emerging from current methodology provide a well-grounded framework for building a clinically credible injury management system within a student sports portal.

2.3 STANDARDIZED DATA FRAMEWORKS AND ANALYTICS FOR SPORTS INJURY ADMINISTRATION

Beyond individual-level injury recording, the aggregation and analysis of injury data across an athlete population serves a distinct and equally important function in institutional sports health management. When injury records are structured consistently and made queryable across variables such as sport, body part, injury type, and time period, administrators gain the capacity to identify systemic risk patterns, allocate medical resources more effectively, and generate evidence-based reports for institutional stakeholders. Realizing this potential, however, requires that the underlying data collection methodology adheres to recognized standards — without which cross-comparison and trend analysis become unreliable.

The IOC consensus statement by Bahr et al. (2020) established internationally recognized methods for recording and reporting epidemiological data on injury and illness in sports, introducing the STROBE-SIIS extension as a standardized framework for sports injury and illness surveillance. The consensus defines the essential data variables that any credible injury surveillance system should capture, including injury type, affected body part, severity, mechanism, and time of occurrence, alongside consistent definitions for what constitutes a recordable injury event. By formalizing these standards, the consensus provides both researchers and system designers with a principled basis for structuring injury databases in ways that support meaningful longitudinal analysis and cross-cohort comparison.

For an institutional sports portal serving a diverse student athlete population, grounding the data model in the IOC's recommended variables ensures that the system captures injury information in a form that is analytically meaningful and consistent over time. An administrative dashboard built on this foundation — one that enables filtering by sport, gender, age group, body part, and date range, and visualizes injury trends across monthly, quarterly, and yearly periods — translates the epidemiological principles of the consensus into practical administrative intelligence. The ability to generate downloadable reports from filtered views further supports institutional reporting requirements, transforming raw injury records into structured evidence for programme evaluation and health planning.

The consensus statement does acknowledge inherent challenges in universal adoption, noting that the diversity of sports settings, resource levels, and existing data infrastructure across organizations makes full standardization difficult in practice. Additionally, as a methodological framework rather than a deployed system, it does not address the specific technical and user experience challenges of implementing these standards within a web-based platform serving non-specialist users such as university administrators. In conclusion, the IOC consensus provides an authoritative and well-recognized framework for structuring sports injury data, and its principles form a sound basis for the design of an analytics-oriented administrative module within the SSP that is both operationally useful and methodologically credible.

2.4 SUMMARY OF LITERATURE REVIEW

Section	Title	Author	Description
Workload Monitoring and Injury Risk Management	Is the ACWR Associated with Risk of Time-Loss Injury in Professional Team Sports? A Systematic Review	Andrade et al. (2020)	• ACWR is a validated metric for identifying elevated injury risk through comparison of acute and chronic workload. • Visual risk thresholds (low/moderate/high) support timely, informed decisions on participant readiness. • Workload spikes relative to a participant's baseline are the strongest predictor of injury risk.

Digital Injury Surveillance and Reporting Systems	Sports Injury Surveillance Systems: A Scoping Review of Practice and Methodologies	Costello et al. (2024)	• Effective surveillance systems support multiple reporting pathways, incorporating both athlete self-reporting and clinician-recorded entries. • Variability in data granularity and reporting frequency across existing systems limits consistency and comparability. • Multi-stakeholder workflows that include review and verification mechanisms improve both data volume and clinical credibility.
Standardized Data Frameworks and Analytics for Sports Injury Administration	IOC Consensus Statement: Methods for Recording and Reporting of Epidemiological Data on Injury and Illness in Sports 2020	Bahr et al. (2020)	• The STROBE-SIIS framework defines standardized variables — injury type, body part, severity, mechanism, and date — essential for credible longitudinal surveillance. • Consistent data structure enables cross-cohort comparison and population-level trend analysis across sport, gender, and age group. • Universal adoption remains challenging due to diversity in organizational resources and existing data infrastructure.

Table 2.1: Summary Table of Literature Review

2.5 EXISTING SYSTEMS COMPARISON

Feature	Workload & ACWR Monitoring	Injury Risk Display	Medical Staff Injury Recording	Athlete Self-Reported Injury	Admin Injury Analytics	Bulk Data Import
Kitman Labs	Automated ACWR from wearable data; targeted at elite professional squads	Risk flags surfaced to coaching staff; not athlete-facing	Clinical injury logging with body part and severity fields	Wellness questionnaires only; no formal injury self-report workflow	Population-level dashboards; for performance directors and sports scientists	API-based integrations; no direct Excel upload
Teamworks	Configurable load monitoring; requires manual setup per organisation	Configurable alerts; relies on staff to interpret and act	Customizable injury forms; supports detailed clinical documentation	Supports athlete input forms; no formal approval workflow	Comprehensive reporting with filters; requires significant configuration	Supports CSV import; no built-in field validation workflow
Catapult Sports	Core strength. GPS wearable-driven ACWR; hardware dependency required	Load spike alerts via wearable dashboard; staff-only view	Not available	Not available	Load trend analytics; no injury-specific population view	Not available
ATS	Not available	Not available	Purpose-built injury tracking with full clinical fields and recovery status	Not available	Basic injury count reports; limited filtering options	Limited import via proprietary format only
SSP 3.0	**Manual activity logging** with automatic ACWR computation; no hardware required	**Athlete-facing risk level** (Low / Moderate / High) displayed directly on personal dashboard	**Structured injury records** with body part, side, type, severity, and recovery status managed by medical staff	**Dedicated self-report pathway** with Pending / Approved / Rejected review states managed by medical staff	**Filterable injury dashboard** by sport, gender, age, body part, and date; with temporal trend charts and PDF report export	**Excel file upload** with column validation, error flagging, and import history log

Table 2.2: Existing Systems Comparison

The comparison above evaluates four established sports health and athlete management platforms — Kitman Labs, Teamworks, Catapult Sports, and ATS — against the proposed SSP 3.0 across six key functional areas. The evaluation reveals that while commercially available systems demonstrate varying degrees of capability, none provides a fully integrated solution that addresses the complete scope of athlete health management required in a student sports context.

Kitman Labs and Teamworks offer the broadest coverage among the reviewed systems, supporting both workload monitoring and injury management with administrative analytics. However, both platforms are designed exclusively for elite professional and high-performance sport organisations, making them unsuitable for deployment in a university setting due to their commercial licensing costs and lack of institutional context. Catapult Sports, while strong in workload and ACWR computation, is fundamentally dependent on proprietary GPS wearable hardware and provides no injury recording or clinical management capabilities. ATS addresses the clinical side of injury documentation effectively but offers no workload monitoring functionality, limiting its utility to reactive injury management rather than proactive risk prevention.

Critically, none of the reviewed systems support a formal athlete self-reported injury workflow with a structured medical review and approval process, nor do any provide a bulk data import mechanism with built-in validation — two features central to SSP 3.0's design. This comparison therefore establishes a clear gap in the existing landscape: no current platform combines hardware-free workload monitoring, multi-stakeholder injury management, population-level analytics, and accessibility within a student sports environment. SSP 3.0 is proposed to address this gap by delivering an integrated, open-access platform purpose-built for university student athletes and their supporting medical and administrative staff.

2.5.1 Kitman Labs

Feature	Workload & ACWR Monitoring	Injury Risk Display	Medical Staff Injury Recording	Athlete Self-Reported Injury	Admin Injury Analytics	Bulk Data Import
Kitman Labs	Automated ACWR from wearable data; targeted at elite professional squads	Risk flags surfaced to coaching staff; not athlete-facing	Clinical injury logging with body part and severity fields	Wellness questionnaires only; no formal injury self-report workflow	Population-level dashboards; for performance directors and sports scientists	API-based integrations; no direct Excel upload

Table 2.3: Kitman Labs

https://www.kitmanlabs.com

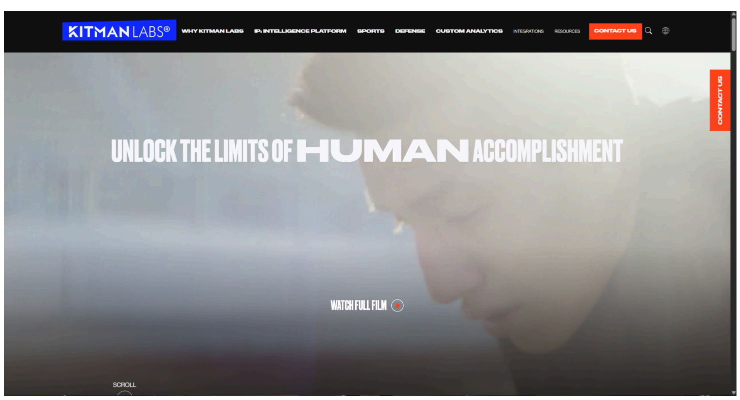

Figure 2.1: Kitman Labs

2.5.2 Teamworks

Feature	Workload & ACWR Monitoring	Injury Risk Display	Medical Staff Injury Recording	Athlete Self-Reported Injury	Admin Injury Analytics	Bulk Data Import
Teamworks	Configurable load monitoring; requires manual setup per organisation	Configurable alerts; relies on staff to interpret and act	Customizable injury forms; supports detailed clinical documentation	Supports athlete input forms; no formal approval workflow	Comprehensive reporting with filters; requires significant configuration	Supports CSV import; no built-in field validation workflow

Table 2.4: Teamworks

https://teamworks.com/ams/

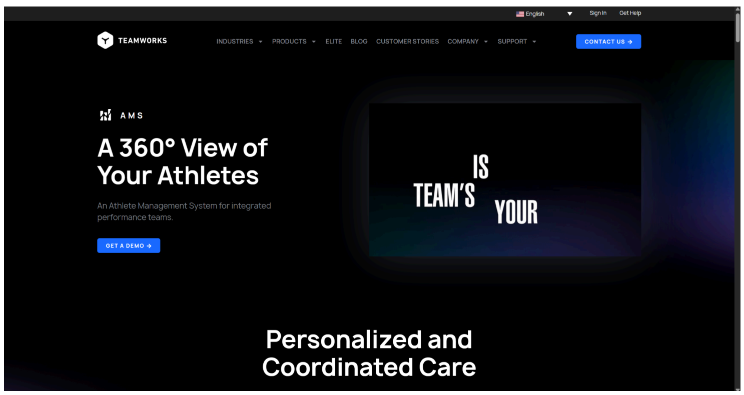

Figure 2.2: Teamworks

2.5.3 Catapult Sports

Feature	Workload & ACWR Monitoring	Injury Risk Display	Medical Staff Injury Recording	Athlete Self-Reported Injury	Admin Injury Analytics	Bulk Data Import
Catapult Sports	Core strength. GPS wearable-driven ACWR; hardware dependency required	Load spike alerts via wearable dashboard; staff-only view	Not available	Not available	Load trend analytics; no injury-specific population view	Not available

Table 2.5: Catapult Sports

https://www.catapult.com

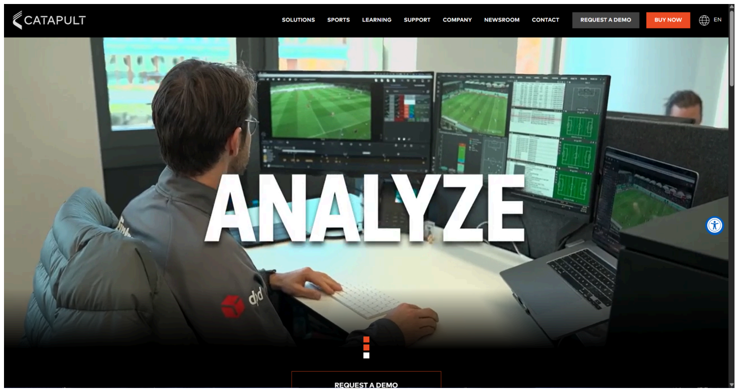

Figure 2.3: Catapult Sports

2.5.4 Athletic Trainer System (ATS)

Feature	Workload & ACWR Monitoring	Injury Risk Display	Medical Staff Injury Recording	Athlete Self-Reported Injury	Admin Injury Analytics	Bulk Data Import
ATS	Not available	Not available	Purpose-built injury tracking with full clinical fields and recovery status	Not available	Basic injury count reports; limited filtering options	Limited import via proprietary format only

Table 2.6: ATS

https://www.athletictrainersystem.com

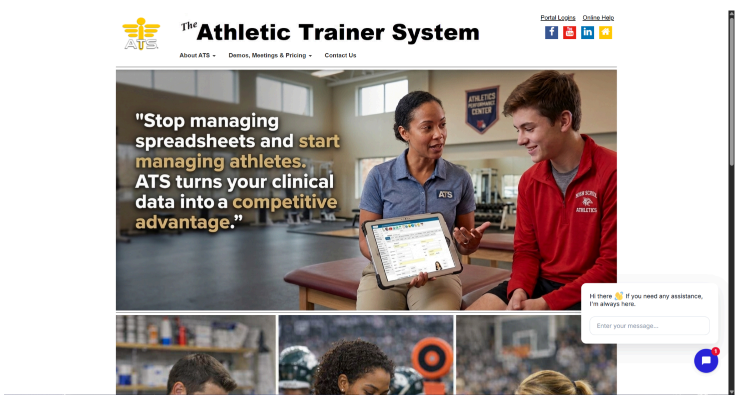

Figure 2.4: ATS

CHAPTER 3: METHODOLOGY

3.1 AGILE METHODOLOGY

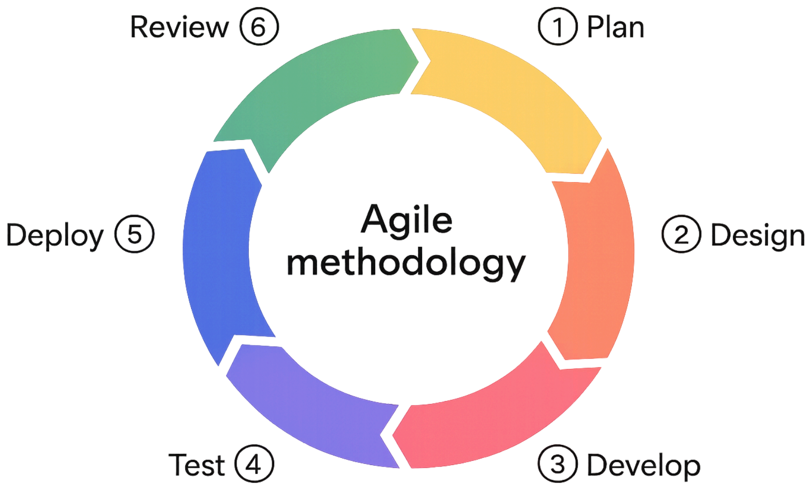

Figure 3.1: Agile Methodology

The chosen methodology for the Athlete Injury Risk Management System is the Agile model. Agile is an iterative and incremental approach to software development that prioritizes continuous stakeholder collaboration, adaptability to evolving requirements, and the delivery of working software in incremental cycles. It is characterized by its flexibility in accommodating changes throughout the development lifecycle, making it well-suited for projects where requirements are expected to be refined progressively based on feedback and new information rather than defined entirely upfront. This project operates in a context that aligns closely with Agile's core principles, as the primary data source for the system's analytics modules was still undergoing cleaning and consolidation at the time development began, and stakeholder consultations were planned as an ongoing process throughout the development period rather than a single upfront requirements gathering exercise.

The Agile process for this project is structured into six phases: Planning, Design, Development, Testing, Deployment, and Review. Each phase feeds into the next while remaining open to iteration based on feedback received from stakeholders and supervisors throughout the project.

3.1.1 Planning

The planning phase establishes the foundation for the entire system. The main goal of this phase is to understand the user requirements, define the problem statements, and establish specific project objectives. This phase involves heavy interaction with ISN as the collaborating stakeholder in order to determine the functional and non-functional requirements of the project. The process includes conducting interviews with ISN representatives to gather insights into their expectations and current workflow challenges. Through this process, the limitations of the existing system are identified and the collaborator's most pressing concerns are determined and prioritized. The requirements gathered during this phase form the product backlog from which all sprint deliverables are derived.

3.1.2 Design

The design phase focuses on translating the user requirements identified during planning into technical and visual blueprints. During this phase, system analysis and design work is carried out, aided by a suite of UML diagrams. The Entity Relationship Diagram is used to design the database schema, grounding the data model in the IOC-recommended variables for sports injury surveillance. The Use Case Diagram is used to define the system's personas and their interactions with each module. The Activity Diagram is used to map the basic workflows of each feature, including the ACWR computation flow, the self-reported injury approval workflow, and the data import validation flow. In parallel, UI/UX wireframes and mockups are produced in Figma for all six modules, adopting the design language of the existing SSP to ensure visual consistency across both deployed systems.

3.1.3 Development

In the development phase, the designs are executed by building a functional software system. Development is carried out iteratively in sprints, where the development cycle is divided into weekly or bi-weekly cycles and each sprint focuses on one module or a small set of related features. The project is structured into two broad sprints aligned with the academic calendar. Sprint 1, carried out during Semester 1, delivers the General Module, Activity Tracking and Logging Module, Injury and Recovery Logging Module, and a partial implementation of the Dashboard and Workload Module. Sprint 2, carried out during the semester break and Semester 2, delivers the completed Dashboard and Workload Module, Data Management Module, Admin Injury Analytics Dashboard Module, and Medical Staff Dashboard Module, along with full UI implementation and system deployment.

3.1.4 Testing

The testing phase is a continuous and iterative process incorporated into each sprint cycle. Features go through extensive testing steps once they are developed, which include unit testing to evaluate individual components such as the ACWR calculation logic and the self-report approval state machine, integration testing to guarantee smooth interaction between dependent modules such as Activity Tracking and the Dashboard module, and User Acceptance Testing to verify alignment with the functional requirements established with ISN. This multi-layered strategy ensures that every component works properly both individually and in combination with the larger system.

3.1.5 Deployment

In the deployment phase, the system is deployed into a pre-production environment that mirrors the live setup. Demonstrations of the system are conducted with ISN representatives, allowing them to experience the system in a realistic environment. This session is important because it enables the collaborator to validate features, test workflows, and offer constructive feedback for improvements before the system is moved to full production. Deployment is carried out in consultation with ISN's server and data security requirements, given the confidential nature of the athlete data handled by the system.

3.1.6 Review

The review phase concentrates on reflection, feedback, and future planning. This phase includes evaluating the codebase, architecture, and system requirements to verify alignment with system quality standards and project objectives. Feedback gathered from ISN personnel through User Acceptance Testing is analyzed and used to identify areas for improvement and inform recommendations for future development. Given the multi-generation nature of this FYP project, the review phase also produces handover documentation to enable future developers to extend or maintain the system effectively.

3.2 DATA GATHERING METHODOLOGY

During the requirements phase, data was gathered through two complementary methodologies: a stakeholder interview as part of the formal collaboration initiative with ISN, and a literature review to validate and contextualize the requirements identified. Together these two approaches form the basis of the product backlog from which sprint deliverables are derived.

3.2.1 Stakeholder Interview

A meeting was conducted with Dr. Thung from Institut Sukan Negara, facilitated by project supervisor Dr. Hoo Wai Lam. The objective was to gain a comprehensive understanding of ISN's current challenges and specific requirements for an injury risk management platform.

Dr. Thung highlighted that ISN oversees 52 sports across multiple programs and currently lacks a centralized digital system for injury tracking and analytics. He identified two primary user groups with distinct needs. Administrators require a holistic overview of injury data across all sports, filterable by sport, gender, age group, body part, injury type, and time period, with the ability to generate standard monthly reports and customized PDF outputs. Medical staff — including physios, doctors, and strength and conditioning specialists — require the ability to trace individual athletes by IC number, view their full injury history, monitor their development over time, and contextualize their condition against sport-level patterns.

Dr. Thung also emphasized the importance of a simple data upload mechanism that allows new screening data to be added to the system without complex steps, and noted that force plate data may be incorporated as an additional dataset in future iterations. The confidentiality of athlete data was highlighted as a critical requirement, necessitating the signing of a Non-Disclosure Agreement before any data is shared.

Key features identified from the interview:

1. **Activity Tracking and Logging Module** — Athletes should be able to log their own training activity records to enable workload monitoring and injury risk assessment through the ACWR methodology.

2. **Dashboard and Workload Module** — A user-facing dashboard that computes and displays the athlete's current injury risk level based on logged activity, providing a more detailed and data-driven view than the existing health dashboard.

3. **Injury and Recovery Logging Module** — Medical staff should be able to log official injury records against athletes, and athletes should be able to self-report non-training injuries through a formal submission and approval workflow, mirroring the multi-stakeholder surveillance architecture identified as most effective in the literature.

4. **Data Management Module** — A simple Excel upload mechanism that allows screening data to be added to the system and reflected in the dashboards without complex steps.

5. **Admin Injury Analytics Dashboard Module** — A holistic dashboard for administrators showing injury summaries filterable by sport, gender, age group, body part, and time period, with temporal trend charts and PDF report generation, grounded in the IOC-recommended data variables for injury surveillance.

6. **Medical Staff Dashboard Module** — An individual-focused dashboard allowing medical staff to search for a specific athlete and view their full injury history, injury trend, and sport-level context in a single view, supporting longitudinal clinical decision-making.

3.2.2 Literature Review

In addition to the stakeholder interview, a literature review was conducted to validate the requirements identified and establish the theoretical foundation for the system's core features.

The review of workload monitoring literature established the ACWR as a widely validated metric for quantifying injury risk, with visual risk thresholds shown to significantly aid practitioners in making timely decisions about athlete readiness (Andrade et al., 2020). This directly underpins the Dashboard and Workload Module's risk classification logic and justifies the inclusion of activity tracking as a core data input for the system.

The scoping review of digital injury surveillance systems identified multi-stakeholder reporting architectures as the most effective approach for capturing comprehensive injury data, validating the dual-pathway design of the Injury and Recovery Logging Module where athlete self-reports feed into a clinician review and approval process (Costello et al., 2024). The review also highlighted the persistent challenge of underreporting bias in self-reporting systems, which informs the design decision to maintain medical staff oversight over all officially recorded injury entries.

The IOC consensus statement on sports injury surveillance provided the authoritative data framework for structuring the system's injury records, establishing the essential variables — injury type, body part, side, severity, mechanism, and time of occurrence — that the database schema and analytics modules are built around (Bahr et al., 2020). The consensus also validated the administrative dashboard's filtering and temporal trend features as operationally meaningful tools for institutional injury management and policy planning.

Together these three bodies of literature provide a comprehensive and well-grounded theoretical foundation for the system's design, ensuring that each module is justified not only by stakeholder requirements but by established academic evidence in workload management, injury surveillance, and sports data analytics. The insights drawn from the literature review formed a key input to the initial product backlog, complementing the direct requirements gathered from Dr. Thung and ensuring that sprint deliverables are aligned with both operational needs and academic best practice.

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.1 REQUIREMENT ANALYSIS

4.1.1 Functional Requirements

Module	Use Case	Title	Description	User Role
General Module	UC-1	Register Account	Register a new account using email and password	Athlete, Medical Staff, Administrator
	UC-2	Login Account	Login with credentials with session management via cookies	Athlete, Medical Staff, Administrator
	UC-3	Reset Password	Reset password via email token link	Athlete, Medical Staff, Administrator
	UC-4	Manage User Profile	View and update personal profile details including sport, age, gender	Athlete, Medical Staff, Administrator
	UC-5	Role-Based Access Control	Restrict access to features and pages based on assigned user role	System
Activity Tracking & Logging Module	UC-6	Log Activity	Input a new activity entry with details such as type, duration, intensity, and date	Athlete
	UC-7	View Activity History	View a list of past logged activities filterable by type and date range	Athlete
	UC-8	Edit Activity	Update the details of a previously logged activity entry	Athlete
	UC-9	Delete Activity	Remove a previously logged activity entry	Athlete
	UC-10	Summarize Activity Data	Automatically compute total weekly activity load from logged entries to feed into the Dashboard module	System

Athlete Dashboard / Workload Module	UC-11	Calculate Workload	Compute the acute (1-week) and chronic (4-week rolling average) activity load from logged entries to derive the ACWR	System
	UC-12	Display Workload Data	Show the athlete their weekly activity totals and ACWR trend over time using charts	Athlete
	UC-13	Determine Risk Level	Apply rule-based thresholds to the ACWR value to classify current injury risk as Low, Moderate, or High	System
	UC-14	Display Risk Level	Show the current risk level to the athlete using a clear visual indicator with a brief explanation	Athlete
	UC-15	Generate Risk Alert	Display a prominent warning on the dashboard when the athlete's risk level is Moderate or High	System
Injury & Recovery Logging Module	UC-16	Log Official Injury Record	Medical staff record an official injury against an athlete including body part, side, injury type, severity, and date	Medical Staff
	UC-17	Update Injury Status	Update the recovery status of a logged injury entry such as Recovering or Recovered	Medical Staff
	UC-18	View Athlete Injury Records	View all official injury records associated with a specific athlete	Medical Staff, Administrator
	UC-19	Delete Injury Record	Remove an official injury record	Medical Staff, Administrator
	UC-20	Submit Self-Reported Injury	Athlete submits a non-training injury report with description and supporting details placed in Pending state awaiting medical staff review	Athlete
	UC-21	Review Self-Reported Injury	Medical staff review a pending athlete-submitted injury report and either approve it into the official record or reject it with a note	Medical Staff

	UC-22	View Self-Report Status	Athlete views the status of their submitted injury reports (Pending, Approved, Rejected) and any notes attached	Athlete
Data Management Module	UC-23	Import Data	Upload an Excel file containing screening data with column validation before committing to the database	Medical Staff, Administrator
	UC-24	Validate Import Data	System checks uploaded file for missing fields, incorrect formats, and duplicate records before committing	System
	UC-25	View Import History	View a log of past data imports including file name, date, number of records added, and any errors encountered	Administrator
	UC-26	Delete Import Record	Remove a previously imported dataset from the system	Administrator
Admin Injury Analytics Dashboard Module	UC-27	View Injury Overview	Display a holistic summary of all injury records across the system including total cases, injury type distribution, and body part breakdown	Administrator
	UC-28	Filter Injury Data	Filter the injury overview by sport, gender, age group, body part, injury type, and date range	Administrator
	UC-29	View Temporal Trends	Display injury trends over time monthly, quarterly, and yearly to identify peak risk periods	Administrator
	UC-30	Generate PDF Report	Generate a downloadable PDF report based on currently applied filters and displayed data	Administrator
Medical Staff Dashboard Module	UC-31	Search Athlete	Look up a specific athlete by name or ID number to access their individual profile	Medical Staff
	UC-32	View Athlete Profile Summary	Display an individual athlete's personal details, sport, age, gender, and physical information in a single view	Medical Staff

	UC-33	View Individual Injury History	Display the full chronological injury history of a specific athlete including type, body part, severity, and recovery status	Medical Staff
	UC-34	View Individual Injury Trend	Display a chart showing how the selected athlete's injury patterns have developed over time	Medical Staff
	UC-35	View Sport-Level Context	Display injury statistics for the athlete's sport alongside the individual view so medical staff can contextualize the athlete against their sport's general patterns	Medical Staff
	UC-36	View Workload History	Display the logged activity and ACWR trend of a specific athlete from the medical staff's perspective	Medical Staff

Table 4.1: Functional Requirements

4.1.2 Non-Functional Requirements

Category	Requirement	Description
Security	Role-Based Access Control (RBAC)	The system shall restrict user actions based on predefined roles such as Athlete, Medical Staff, and Administrator. Each role shall have access only to specific modules and functionalities relevant to their responsibility.
Usability	User-Friendly Interface	The system shall provide a clean and intuitive user interface with consistent layouts, making it easy to navigate across all modules.
	Ease of Use	The application shall be designed so that users can understand and operate it with minimal guidance. There shall be no steep learning curve for new users.

Table 4.2: Non-Functional Requirements

4.2 FUNCTIONAL DECOMPOSITION DIAGRAM

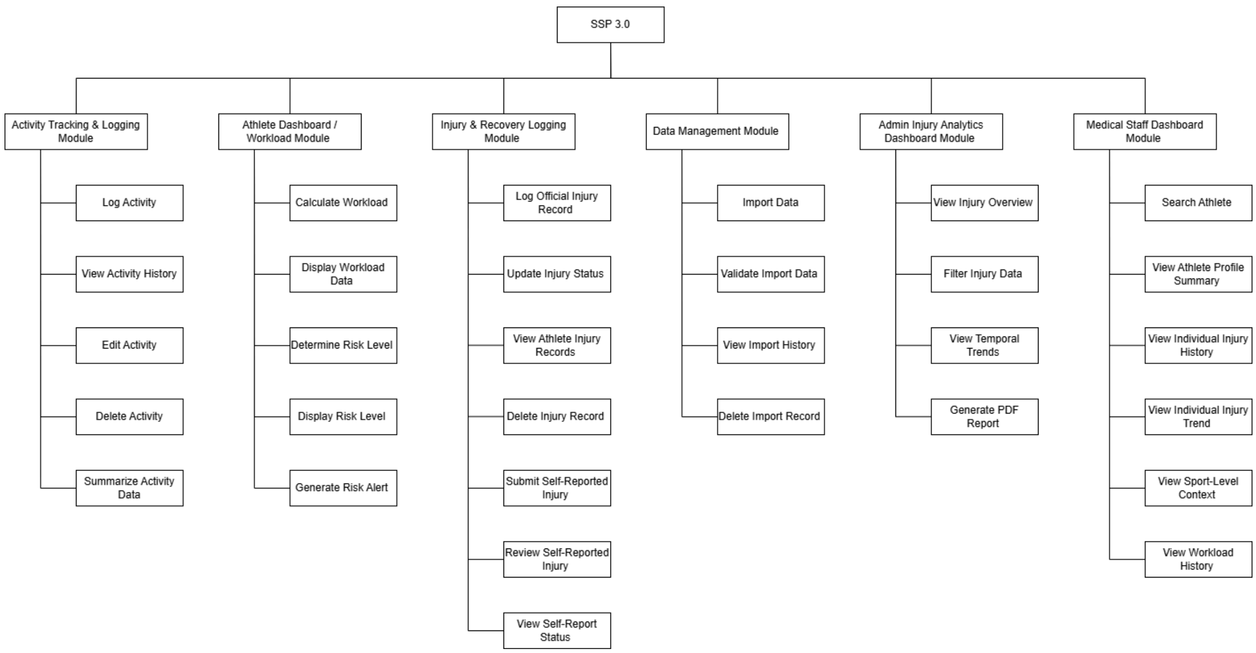

Figure 4.1: Functional Decomposition Diagram

4.3 USE CASE DIAGRAM

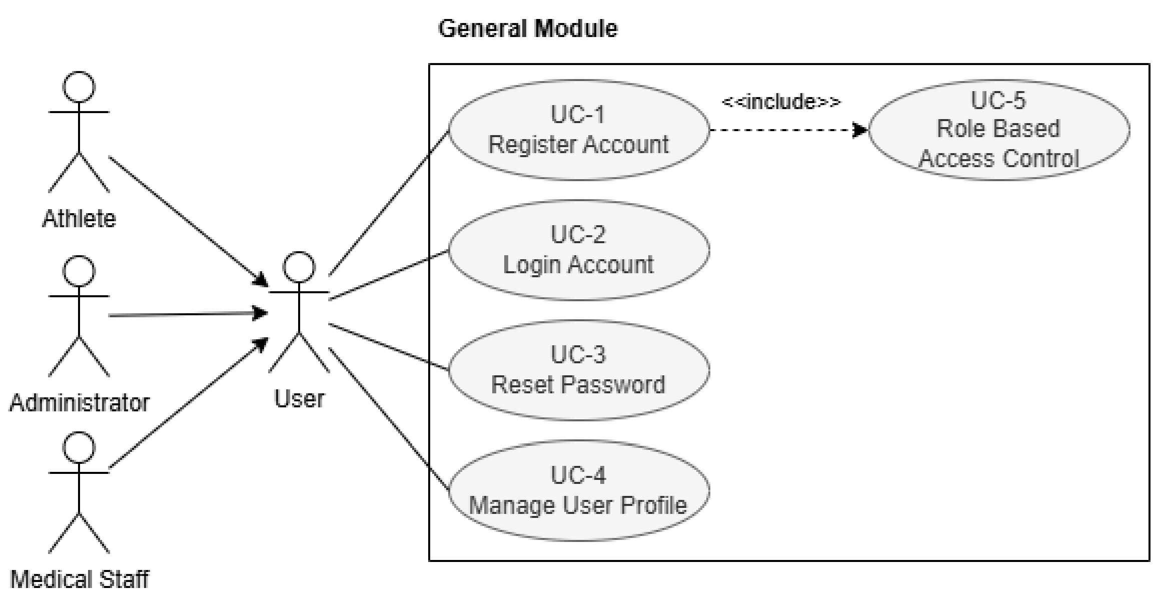

Figure 4.2: Use Case Diagram of General Module

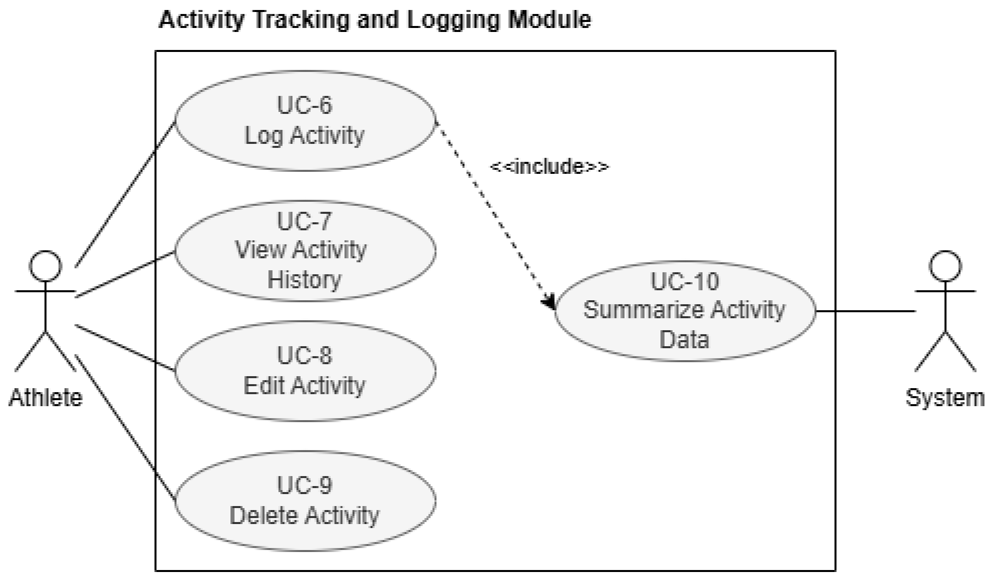

Figure 4.3: Use Case Diagram of Activity Tracking and Logging Module

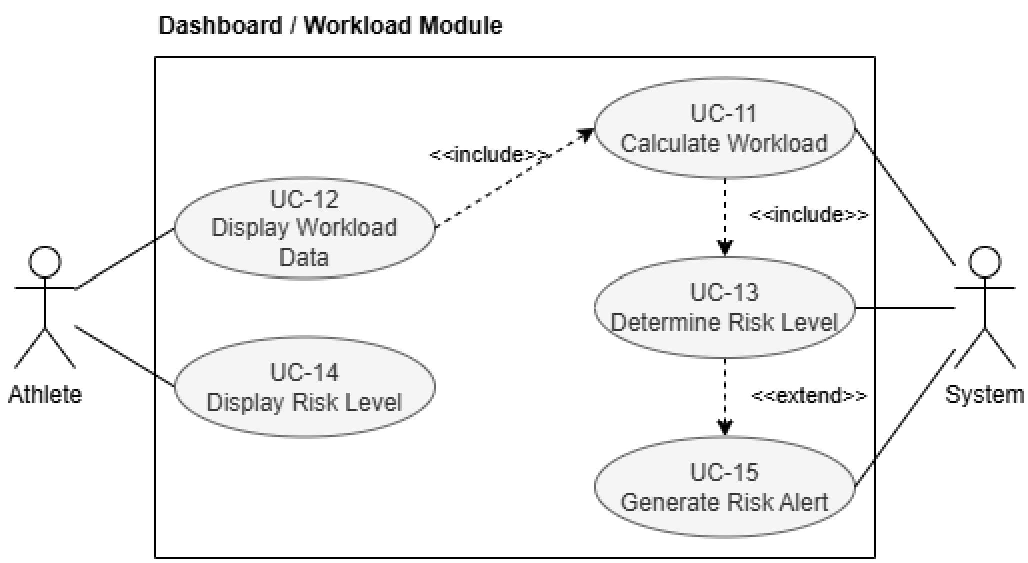

Figure 4.4: Use Case Diagram of Athlete Dashboard / Workload Module

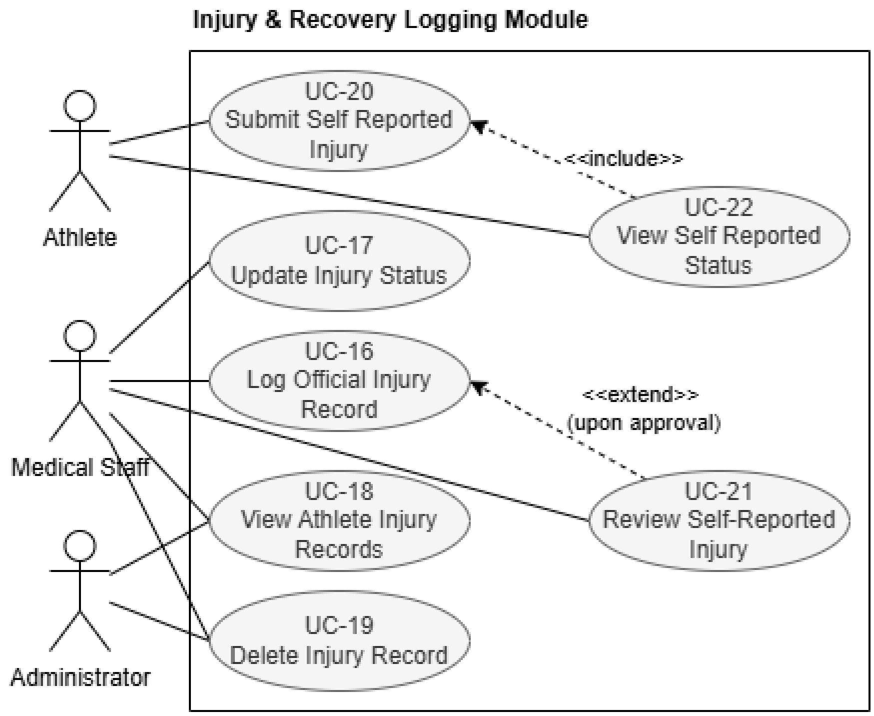

Figure 4.5: Use Case Diagram of Injury & Recovery Logging Module

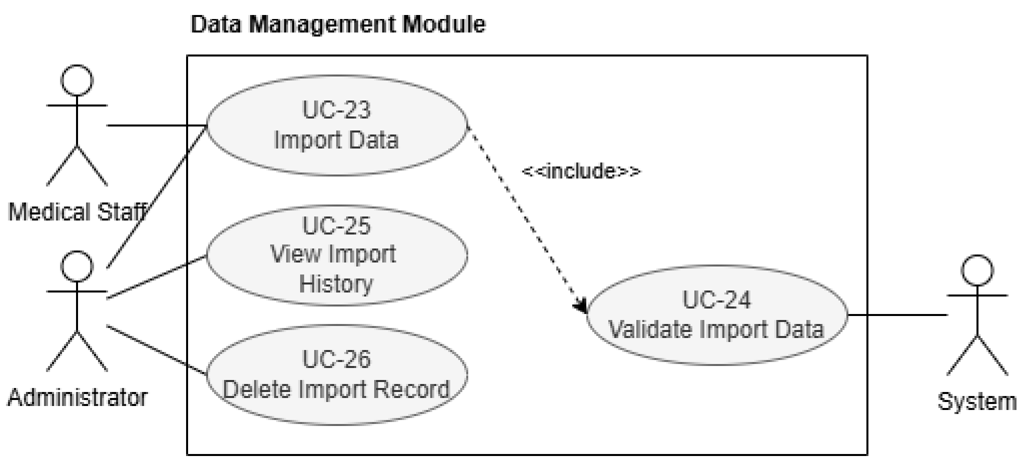

Figure 4.6: Use Case Diagram of Data Management Module

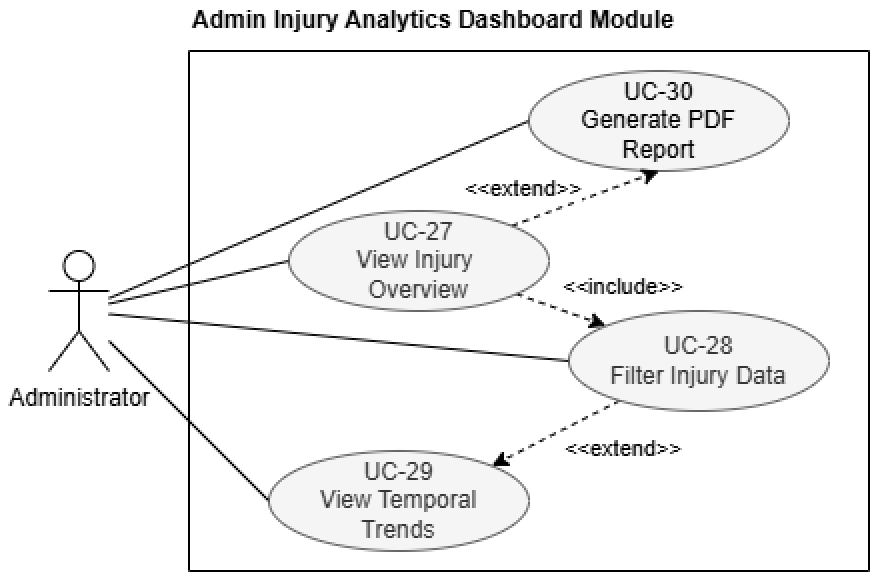

Figure 4.7: Use Case Diagram of Admin Injury Analytics Dashboard Module

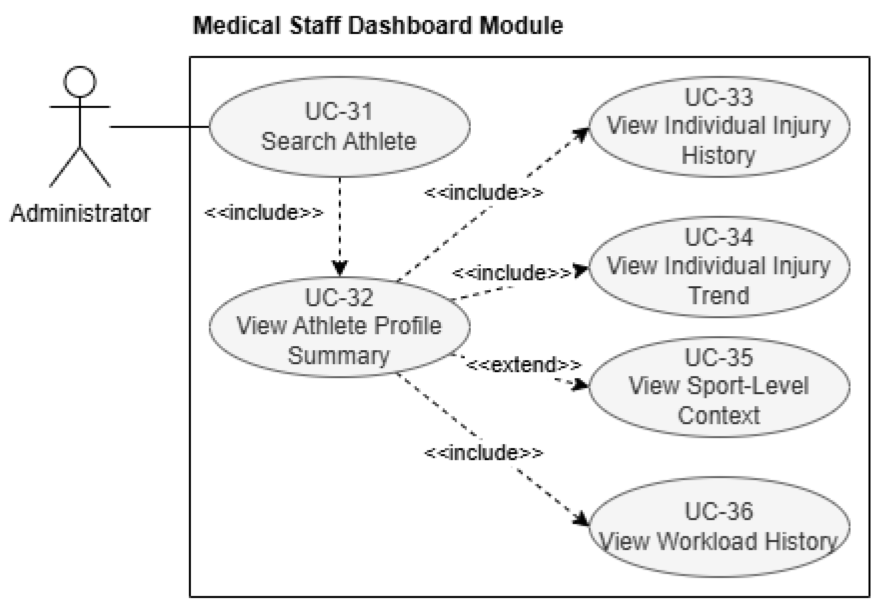

Figure 4.8: Use Case Diagram of Medical Staff Dashboard Module

4.4 ENTITY RELATIONSHIP DIAGRAM

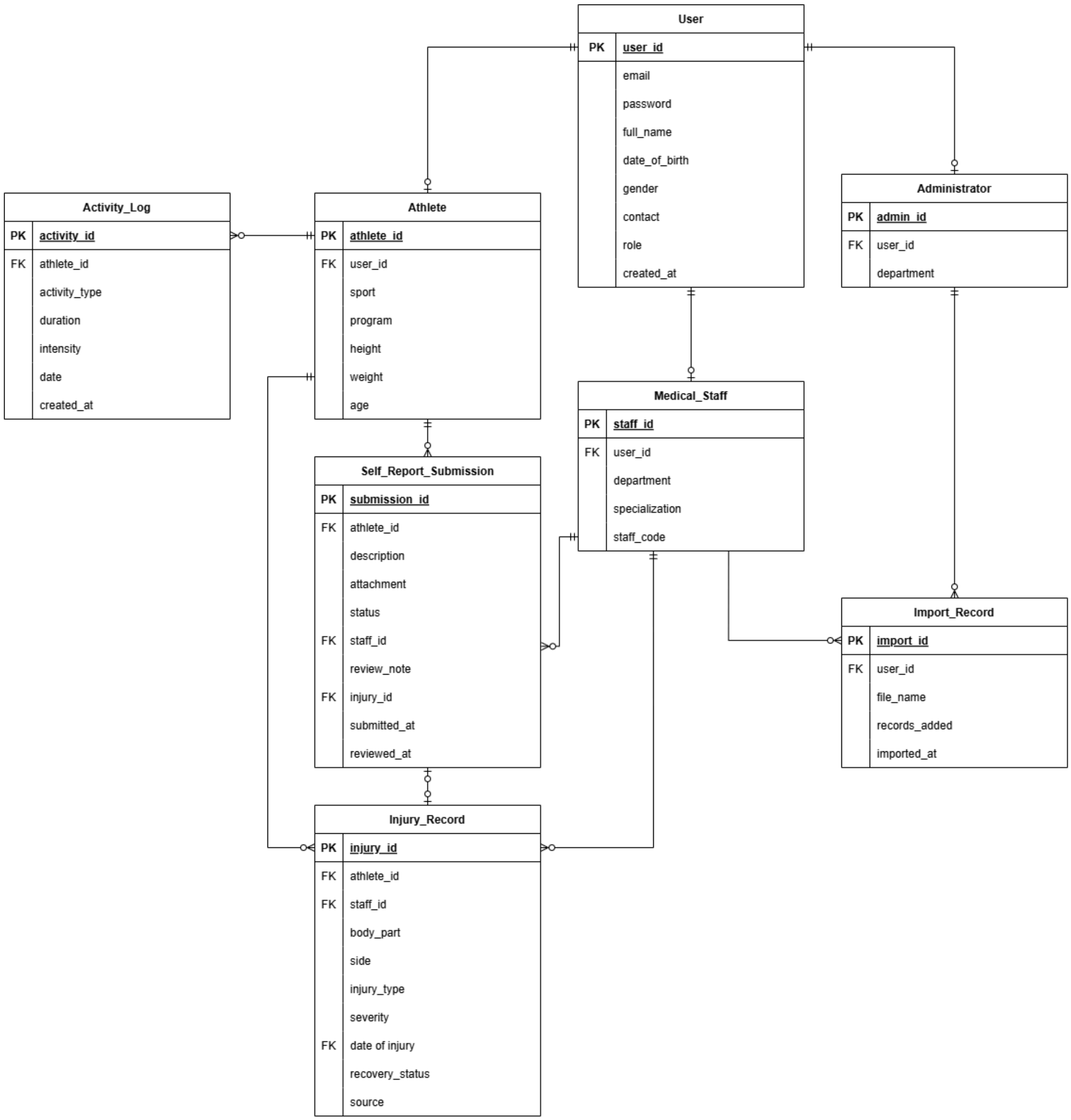

Figure 4.9: Entity Relationship Diagram

4.5 ACTIVITY DIAGRAM

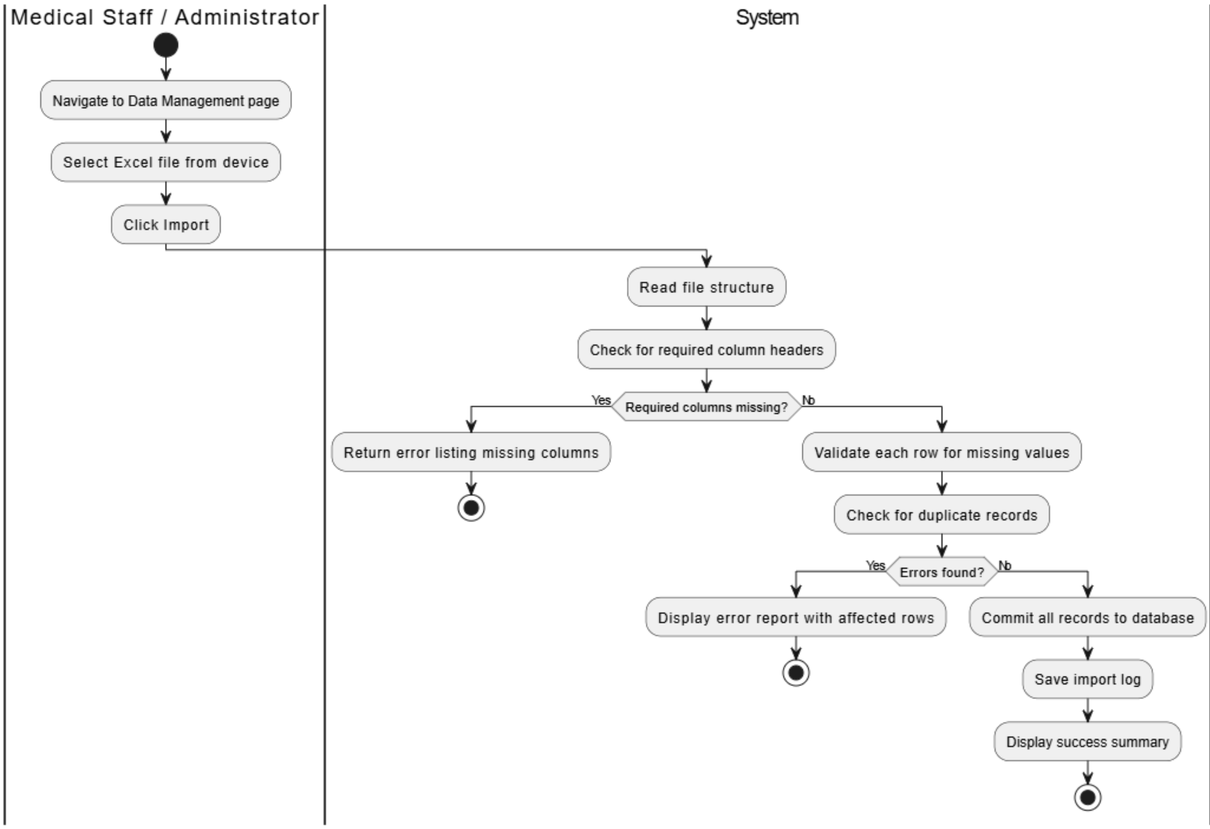

Figure 4.10: Activity Diagram of UC-

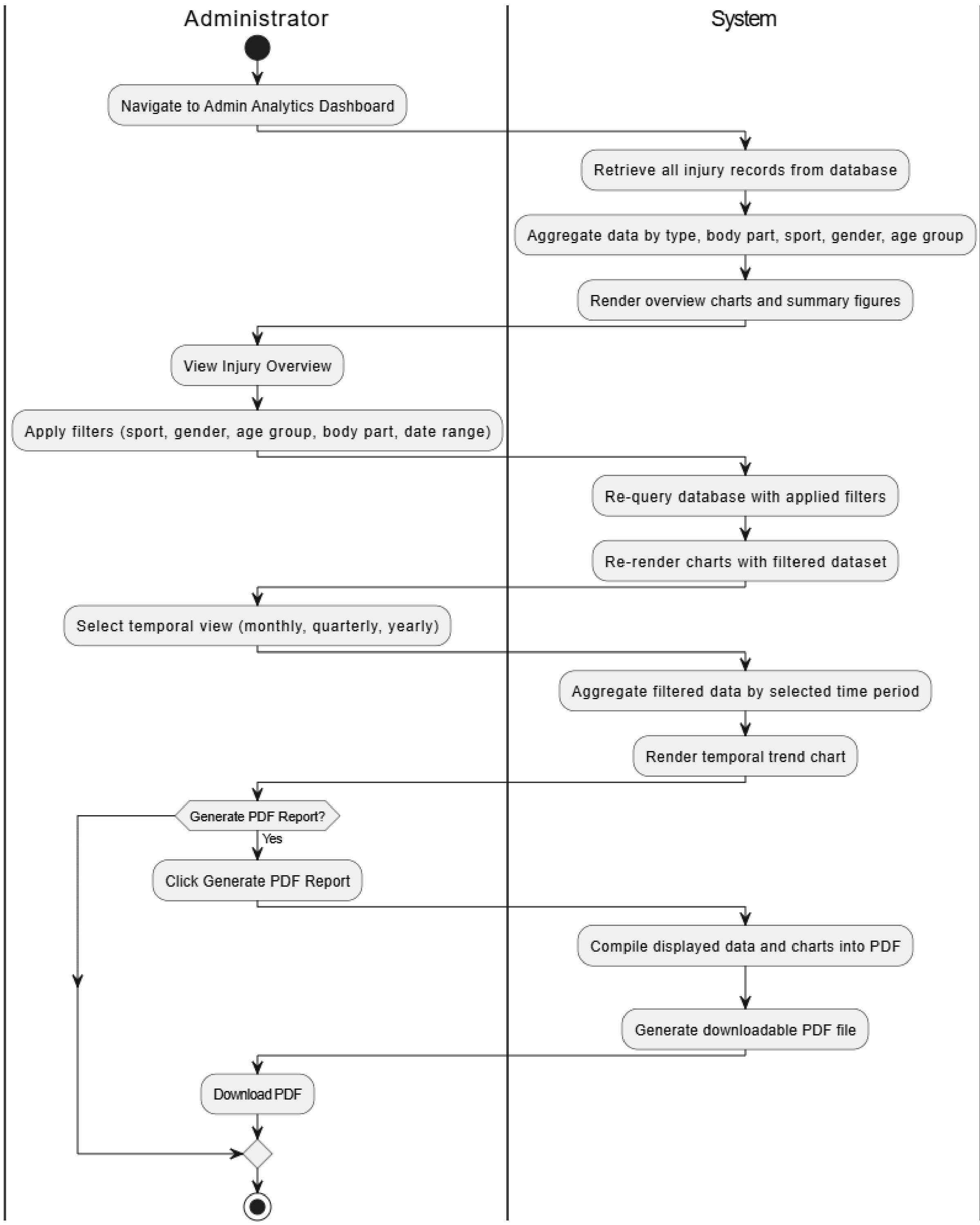

Figure 4.11: Activity Diagram of UC-

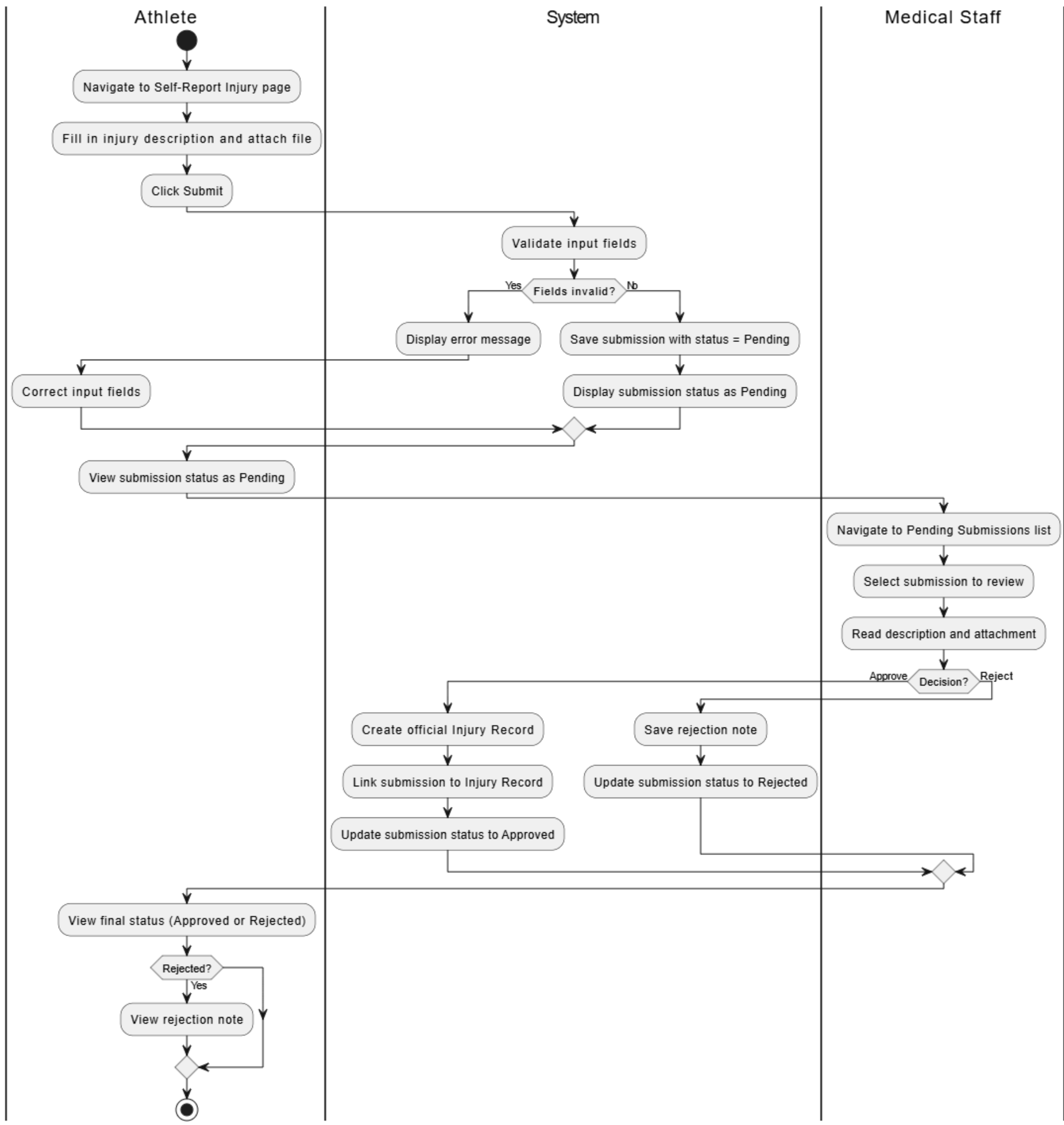

Figure 4.12: Activity Diagram of UC-

4.6 USER INTERFACE DESIGN

CHAPTER 5: SYSTEM DEVELOPMENT

5.1 TECHNICAL IMPLEMENTATION

5.2 USER INTERFACE & USER EXPERIENCE

CHAPTER 6: STAKEHOLDER COLLABORATION

6.1 COLLABORATION LETTER

UNIVERSITI MALAYA

31 MARCH 2026

Dr. Thung Jin Seng
Institut Sukan Negara
Kompleks Sukan Negara
57000 Bukit Jalil
Kuala Lumpur

Sir/ Madam,

Collaborator for Academic Project Courses, Faculty of Computer Science and Information Technology, Universiti Malaya

With reference to the above matter,

2. The Faculty of Computer Science and Information Technology (FCSIT) University Malaya is pleased to invite you to be the collaborator for the **Interactive Student Sports Portal with Health Dashboard.** This project is under the supervision of **Dr. Hoo Wai Lam.**

3. The Academic Project courses are mandatory for final year students of the Faculty of Computer Science and Information Technology as part of their graduation requirements. These courses are designed to nurture academically proficient and technically skilled graduates in the fields of Computer Science and Information Technology. During the course, students actively apply technical expertise, critical thinking, and problem-solving skills to develop innovative products. These projects are intended to meet collaborators' needs and enhance organizational efficiency.

4. This collaboration entails the involvement of the following student(s) from the Faculty of Computer Science and Information Technology working alongside you on the implementation of the aforementioned project. The faculty kindly requests your support in providing the student(s) with necessary information and/or other related resources to facilitate the successful completion of the project.

5. The following are the student(s) who will be involved in this collaboration

Name	**: Lim Jian Chuen**
Matric Number	**: 23005005**
Email	**: 23005005@siswa.um.edu.my**

Faculty of Computer Science and Information Technology
Universiti Malaya, 50603 Kuala Lumpur, MALAYSIA
Tel: (+603) 7967 6300/6301 • http://fsktm.um.edu.my

Figure 6.1: Collaboration Letter (Part 1)

6. We sincerely hope you will give thoughtful consideration to this invitation. We are confident that this collaboration will bring mutual benefits to both the faculty and your organization. Additionally, we kindly request your written consent to formalize your participation in this initiative. Should you require any further information, please do not hesitate to contact the project supervisor, **Dr, Hoo Wai Lam**, at **wlhoo@um.edu.my**.

Thank you.

Yours sincerely,

Professor Dr. Nor Liyana Mohd Shuib
Deputy Dean (Undergraduate)
Faculty of Computer Science and Information Technology
Universiti Malaya
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Figure 6.2: Collaboration Letter (Part 2)

Acceptance of Collaboration

I hereby acknowledge that I accept this proposal for collaboration with the Faculty of Computer Science & Information Technology, Universiti Malaya pertaining to the research project entitled **Interactive Student Sports Portal with Health Dashboard.**

Collaborator signature

Dr. Thung Jin Seng
Institut Sukan Negara
Kompleks Sukan Negara
57000 Bukit Jalil
Kuala Lumpur
Date: 27 April 2026

Figure 6.3: Collaboration Letter (Part 3)

6.2 STAKEHOLDER MEETING

Figure 6.4: First Stakeholder Meeting

CHAPTER 7: CONCLUSION

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APPENDIX